Embedded Systems - Project Specifications

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1. System Level Block Diagram

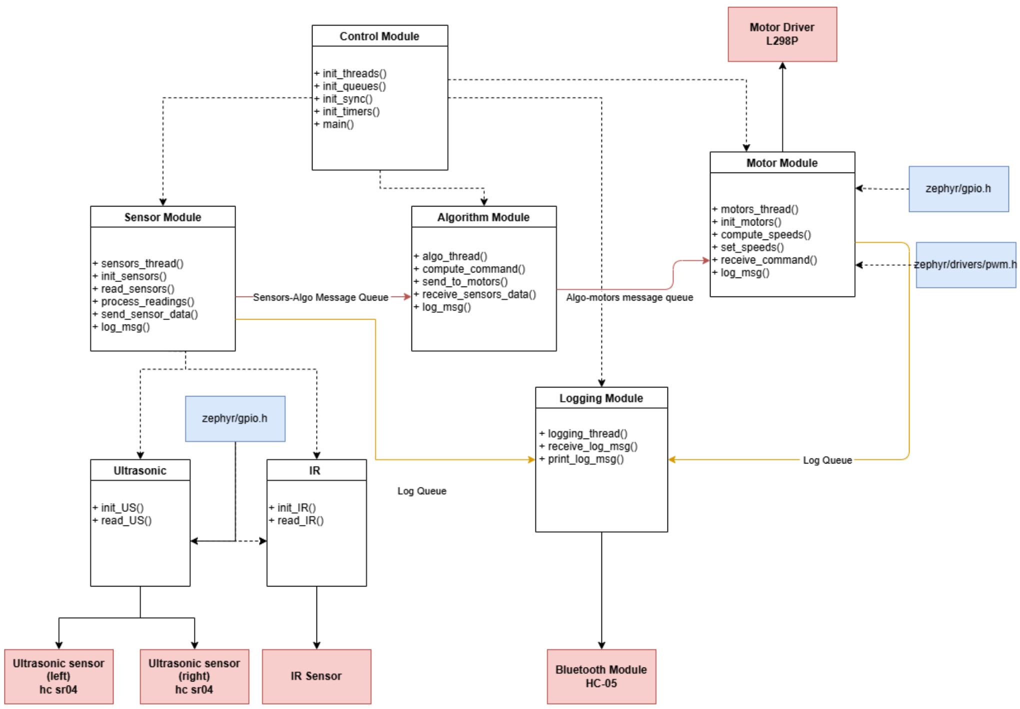

Figure 1: **System Level Block Diagram**

The system-level block diagram illustrates how the different modules in the Micromouse robot interact using message queues (colored lines) and connect to relevant hardware (in red) and to associated Zephyr dependencies (not fully listed, in blue). Below is a brief overview:

1. **Control Module:** Initializes all threads, synchronization primitives, and message queues.
2. **Sensor Module:** Reads data from the IR and Ultrasonic sensors, processes it, and sends it to the Algorithm module via the ”Sensors-Algo” message queue.
3. **Algorithm Module:** Determines the robot’s action (turn left, right, move forward, etc.) and sends movement commands to the Motor module via the ”Algo-Motors” message queue.
4. **Motor Module:** Adjusts motor speeds based on the received commands.
5. **Logging Module:** Receives log messages from other modules via the ”Log” message queue and transmits them via Bluetooth.

2. Thread Flowcharts

Each of the four primary modules in the system corresponds to a separate thread. Their flowcharts below visually outline the core operations of each thread:

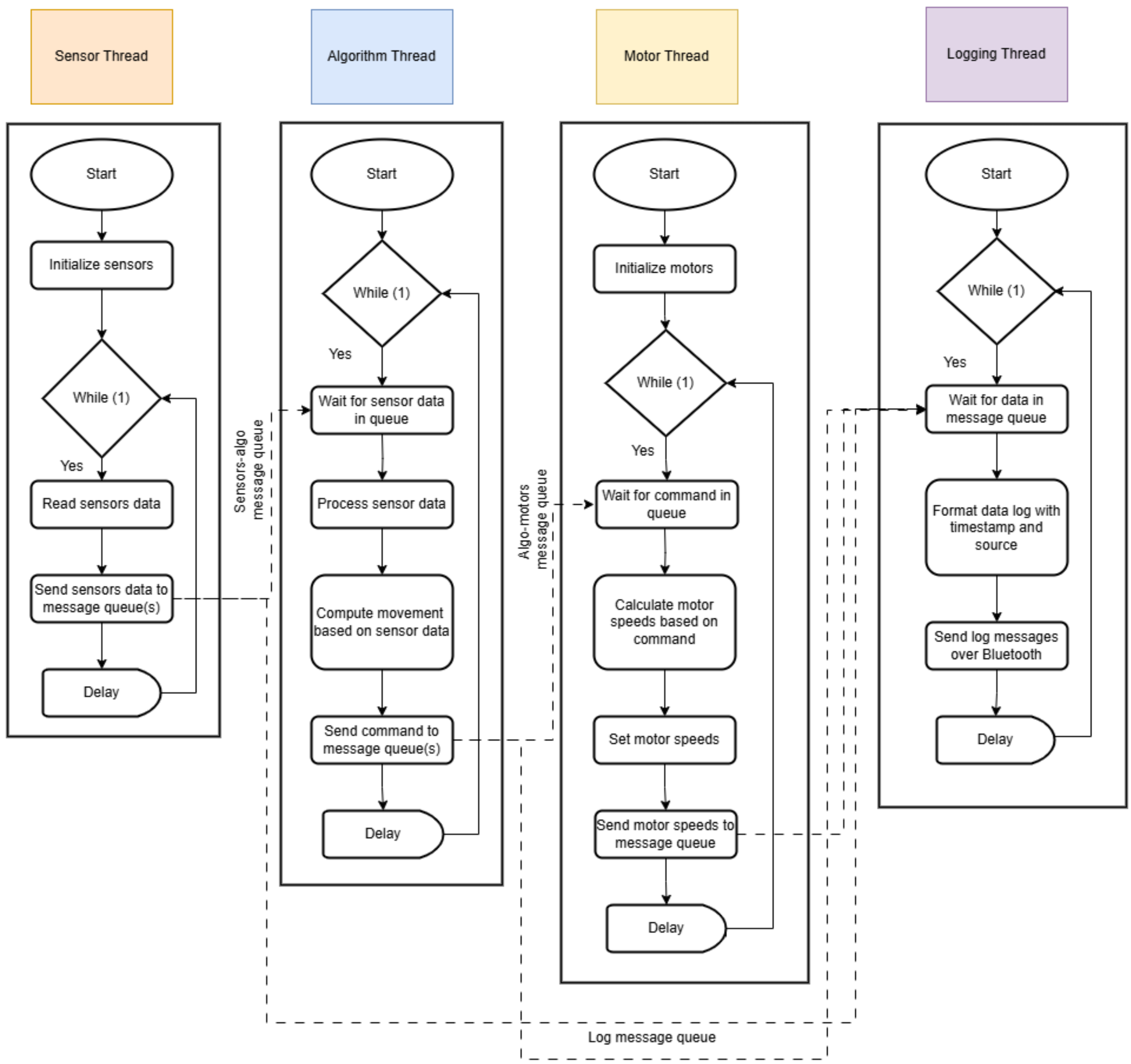

Figure 2: **Thread Flowcharts**